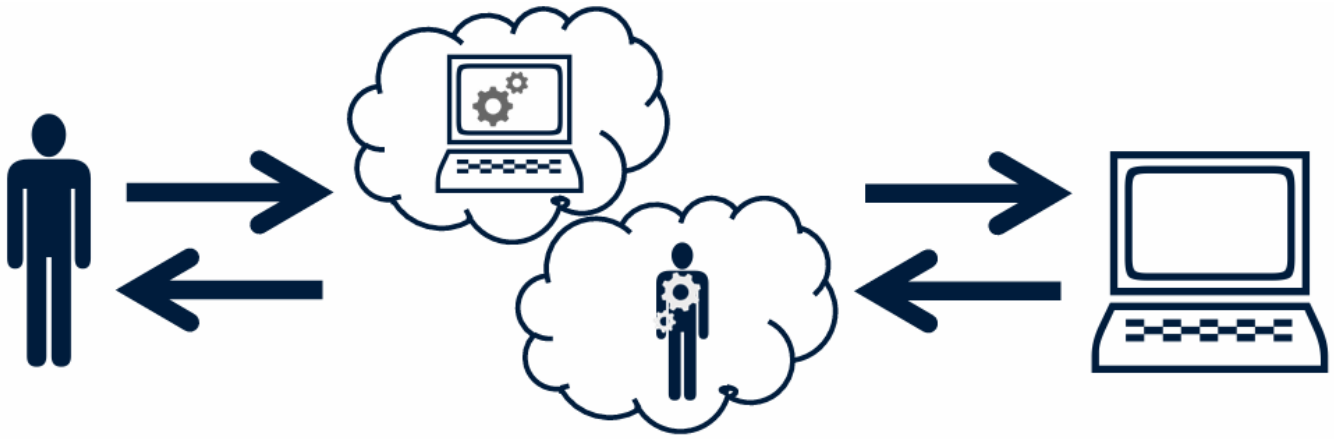
3 – Model Integration

User interaction, Clustering, Dimensionality reduction, Signification object detection, Degree-of-interest functions, Explainability

Human-Computer Interaction

- Interaction is the communication between the user and the system



Activities: What the user does to trigger a change in the computer (*Action*)

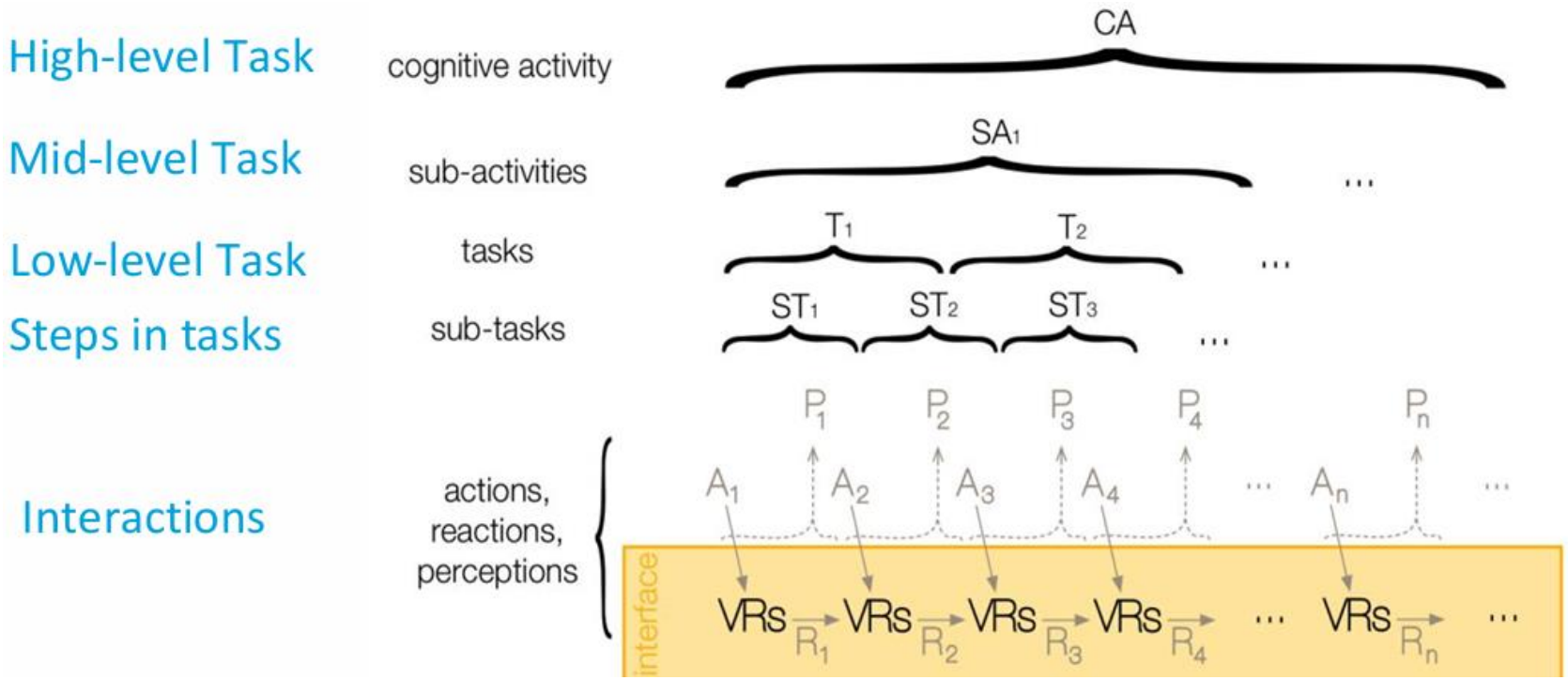
Metaphor: What the user thinks the computer is doing and vice versa (*Understanding*)

Architecture: What the computer actually does (*Reaction*)

- Common modalities include mouse, keyboard, and touch

Cost of Interaction

- Every interaction has a cost:
  - (High) **Mental Cost**: keeping track of the state, planning how to achieve goal, etc.
  - (High) **Physical Cost**: precise mouse movement, deep menus, long eye movements, etc.



Cost of Interaction

- **Indirect Interaction**: interact using separate elements (buttons, sliders, etc.)

- The cost of interaction can be measured in multiple ways:
  - **Spatial Separation**: controls are not positioned where we are looking
  - **Temporal Separation**: time it takes for interactions to be processed
  - **Conceptual Separation**: expectation does not align with implementation

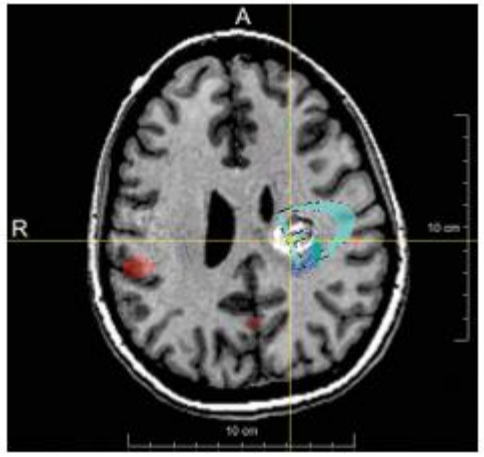
- **Direct interaction** (interact with the visualization) reduces spatial separation
- Loading bars reduce temporal separation
- **Scented Widgets** (show what will happen) reduce conceptual separation

Interaction: View Manipulation

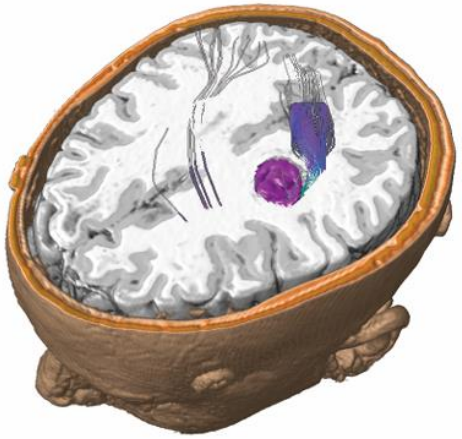
Unconstrained Navigation: camera can move anywhere
Constrained Navigation: some restrictions in place

Geometric Zooming: object stays fixed but scales with distance
Semantic Zooming: object changes depending on distance (e.g. only show label when zoomed in close enough)

- **Selection and Highlighting**: *hover / click / drag*
  - Gain additional information on items
  - Intermediate step for other interactions
- **Navigation**: *zoom / pan*
  - Browse through large amounts of data (overview first, details later)
- **Navigation**: *focus / context*
  - Local distortion (e.g. fisheye)
  - Superimpose details (e.g. windows magnifier)
  - De-emphasize (e.g. filter variables)
- **Arrange**:
  - Sort the order of dimensions / variables



Slice
(1 hyperplane)



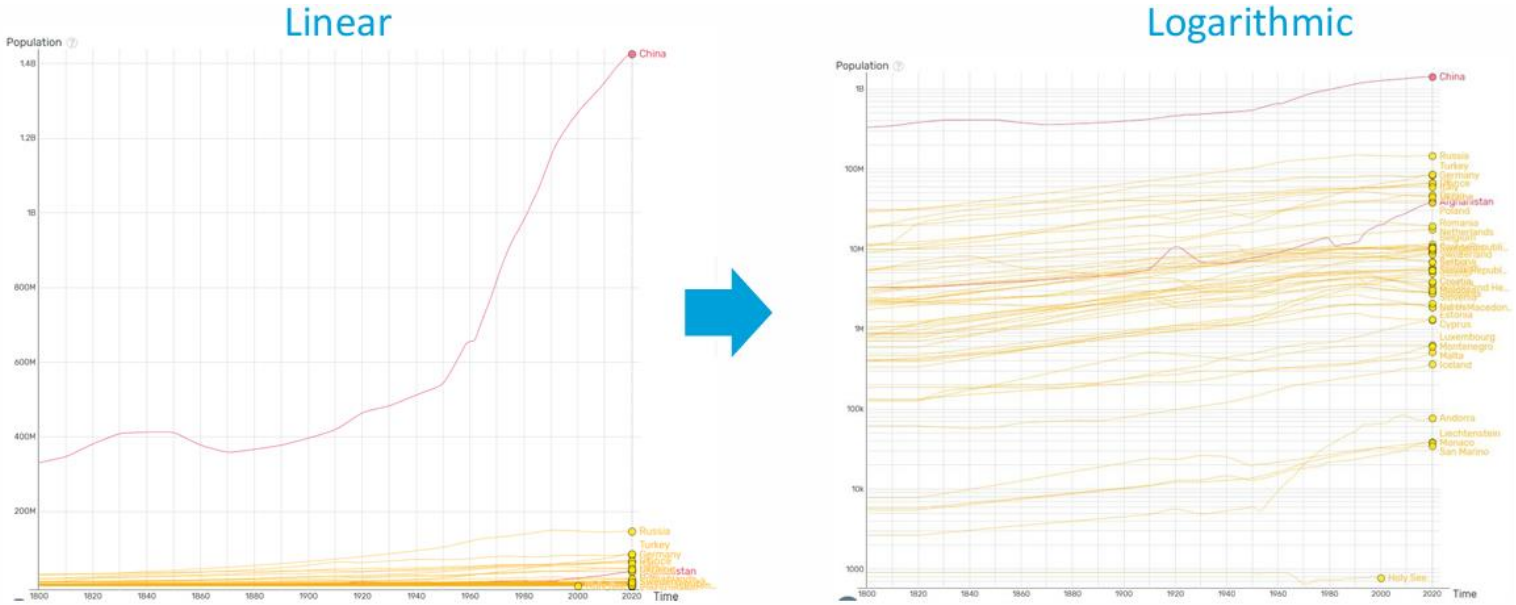
Cut
(filtering)

Interaction: Change the Mapping

- *Add* a new mapping (color, shape, etc.)
- *Change* existing mapping (exploration)

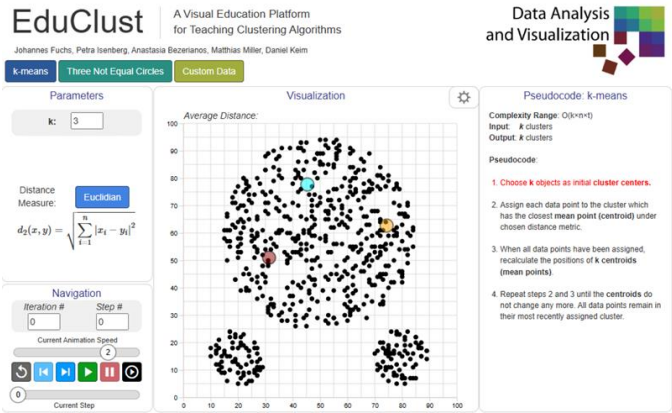
Interaction: Data Manipulation

- **Filtering** (focus on a subset of items)
- **Normalization** (e.g. convert to log scale)
- **Editing / Annotating**



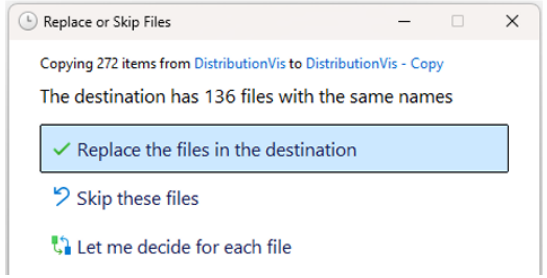
Interaction: Parameter Refinement

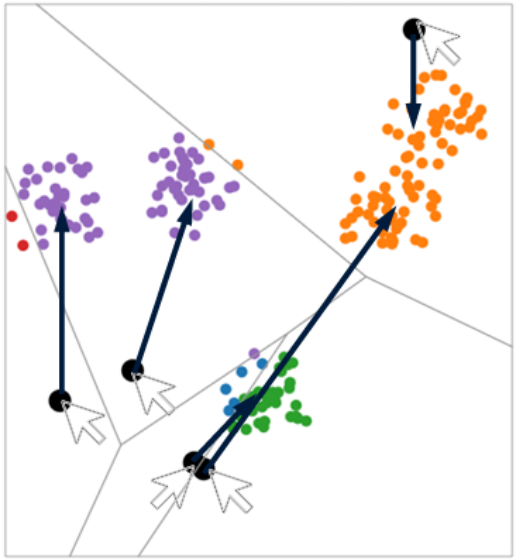
- Interacting with the parameters of the model



Interaction: Model Building

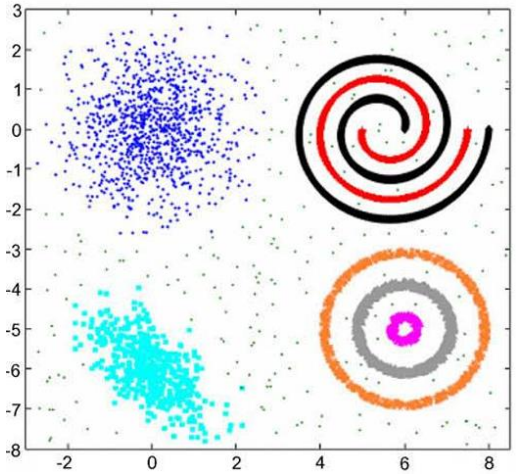
- Directly interacting with the model





Clustering

- **Clustering**: automatically finding meaningful groups of similar objects
- In visual analytics → commonly used for **exploration** and **summarization**



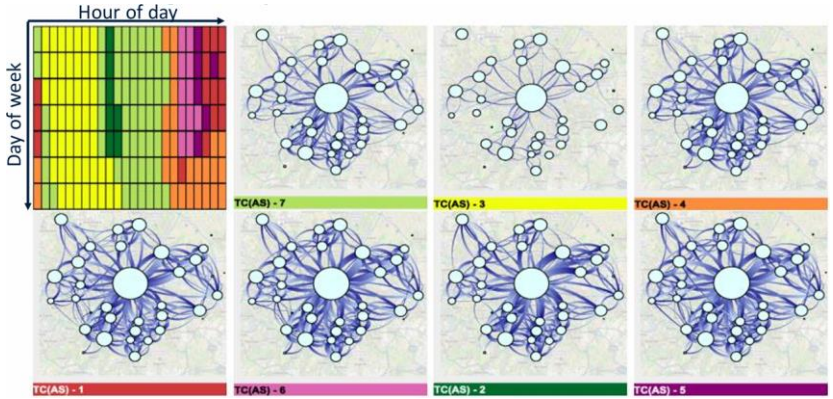
- **Flat Clustering**: each object is in exactly one cluster
- **Hierarchical Clustering**: a set of nested clusters

- **Complete Clustering**: all data points are assigned
- **Partial Clustering**: some data points can be labelled as outliers

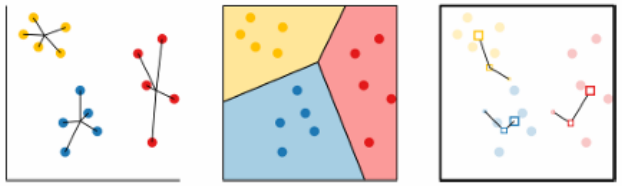
Model Visualization

- **Model Visualization**: using the output / state of a model for visualization

- **Using the model as a tool**: *support* the user by doing something automatically that is difficult to do by hand



- **Understanding the model**: *identify* which aspects need to be explained, build the visualization to support this



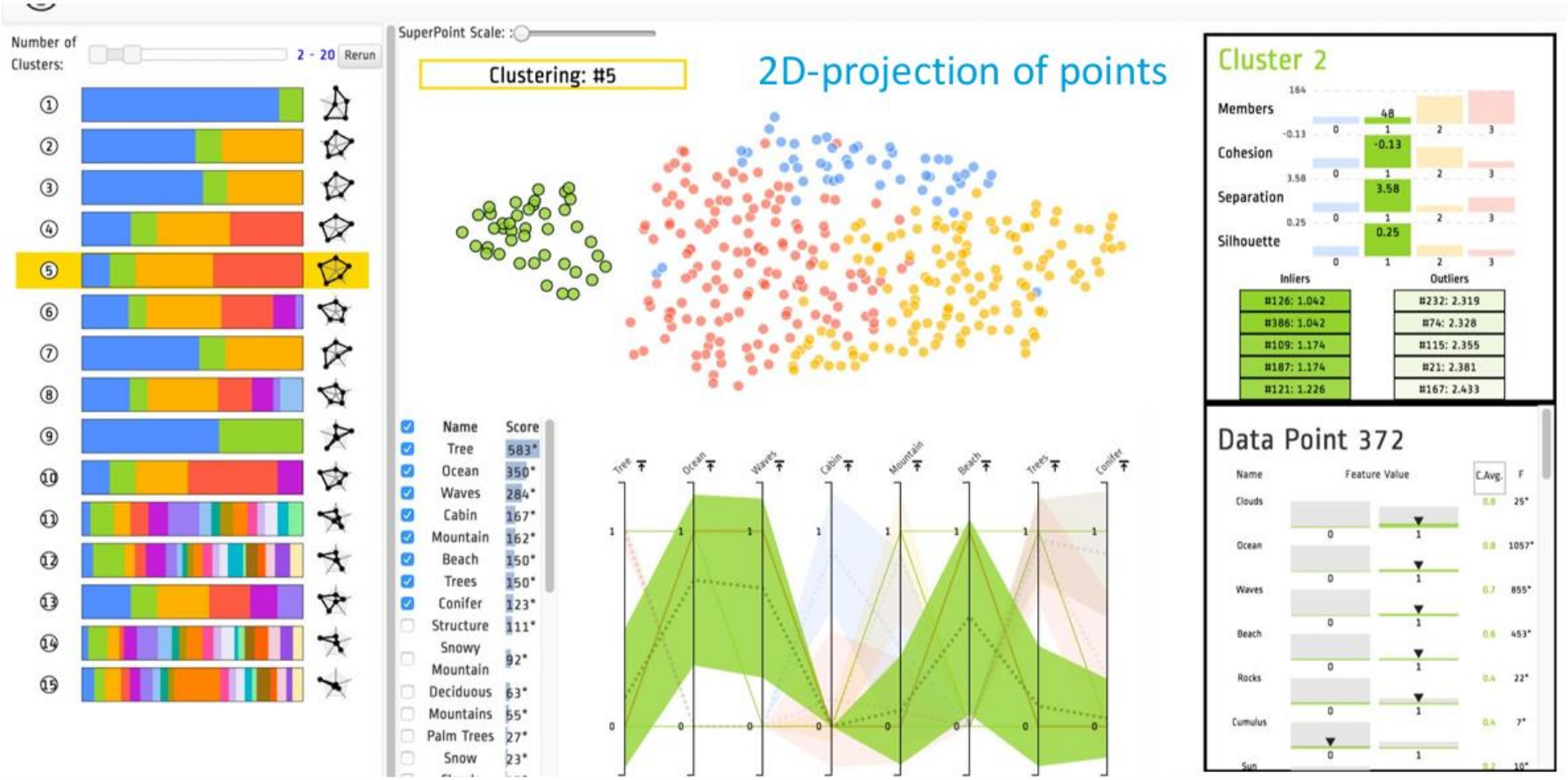
Model Visualization

Kwon, Bum Chul, et al. "Clustervision: Visual supervision of unsupervised clustering." *IEEE transactions on visualization and computer graphics* 24.1 (2017): 142-151.

Clusterings

Radar plot of metrics

Cluster detail



Feature importance score

Parallel Trends (min-max)

Data detail

Dimensionality Reduction

- Reduce the number of dimensions while minimizing loss of information
- **As tool**:
  - Use human vision to solve the visualization task in low-dimensional space
  - Can also map to other visual variables (*exploration*)

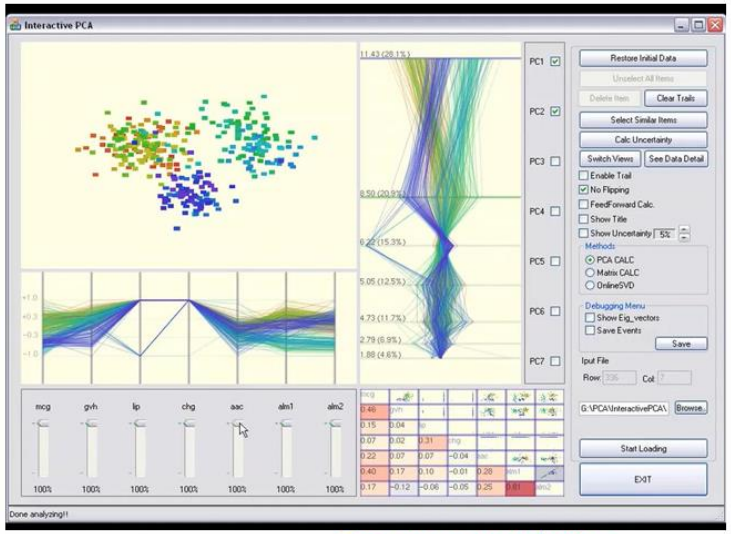
- Understanding the model is the goal

Eigenvectors used for DR

2D Projection (overview)

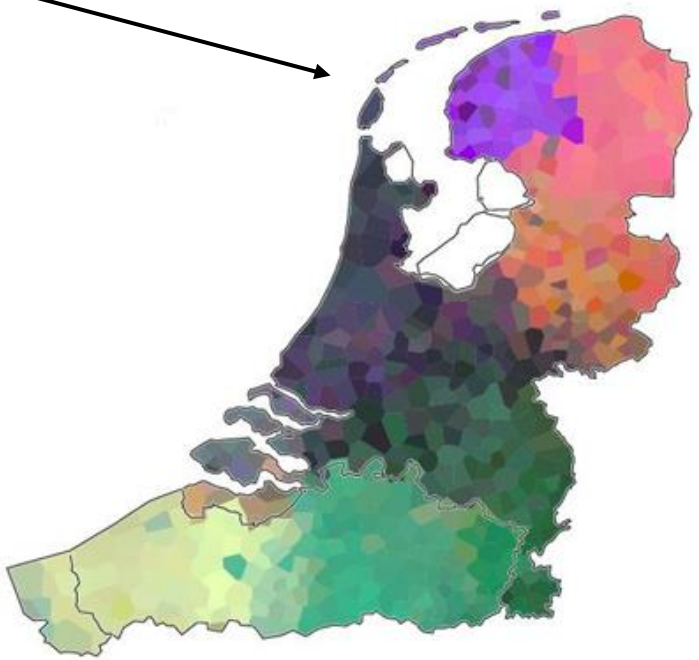
Data view (Detail)

Control of relative importance of each feature for projection



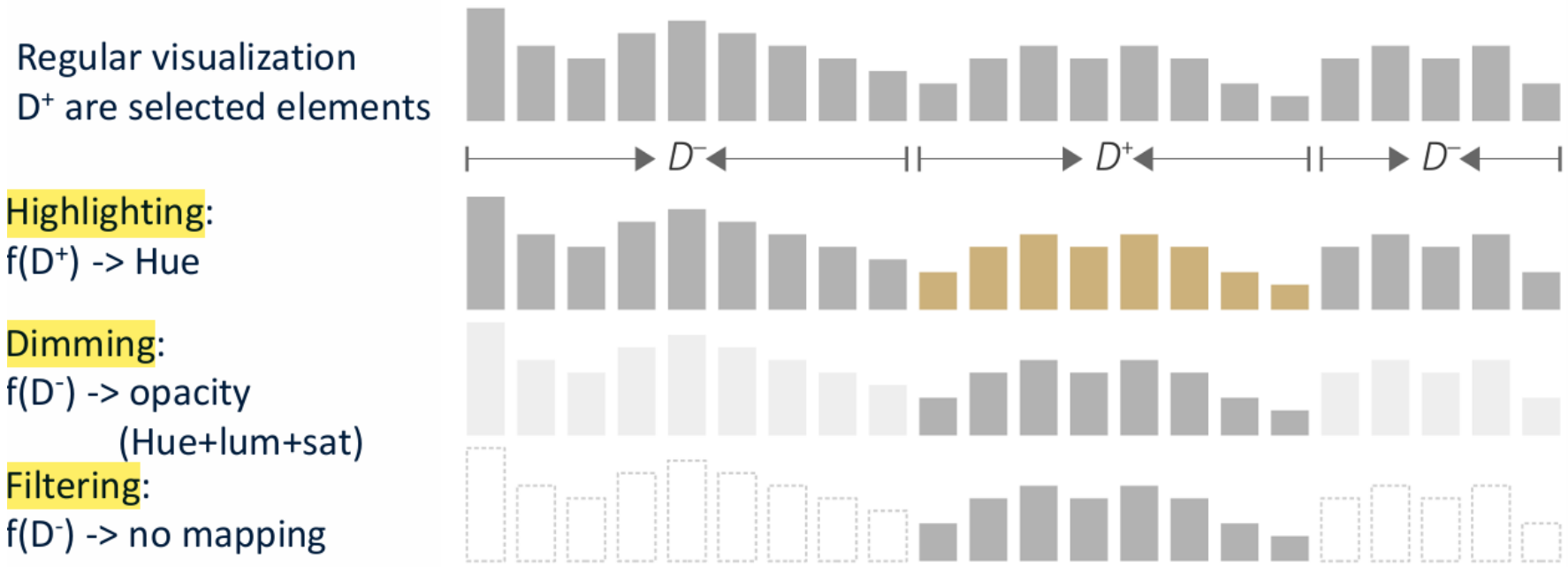
Feature correlation

Meaning of position is technique-dependent!



Significant Object Detection

- **Significant Object Detection**: automatically locate *X*, for example:
  - Locate outliers in back transactions
  - Locate superspreaders of misinformation
  - Locate movies you might like



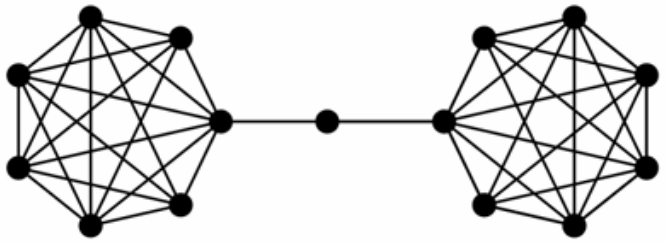
Degree-Of-Interest (DOI)

"Search, Show Context, Expand on Demand"

- *Interest* is relative to the current user and current focus
- Total interest of a point x given a focal point y and search z:

$$DOI(x \mid y, z) = \alpha \cdot API_{diff}(x) + \beta \cdot UI_{diff}(x, z) + \gamma \cdot D(x, y)$$

- API: **A P**riori **I**nterest function (centrality measure, page-rank, etc.)
  - **Degree Centrality**: number of edges connected to this node
  - **Closeness Centrality**: inverse of the sum of shortest distances to all other nodes
  - **Betweenness Centrality**: number of shortest paths that pass through this node
- $diff$: diffusion (take interest of neighbors into account)
- UI: **U**ser **I**nterest, given search z (e.g. similarity of node and search item)
- D: distance to focal point y (e.g. graph distance)

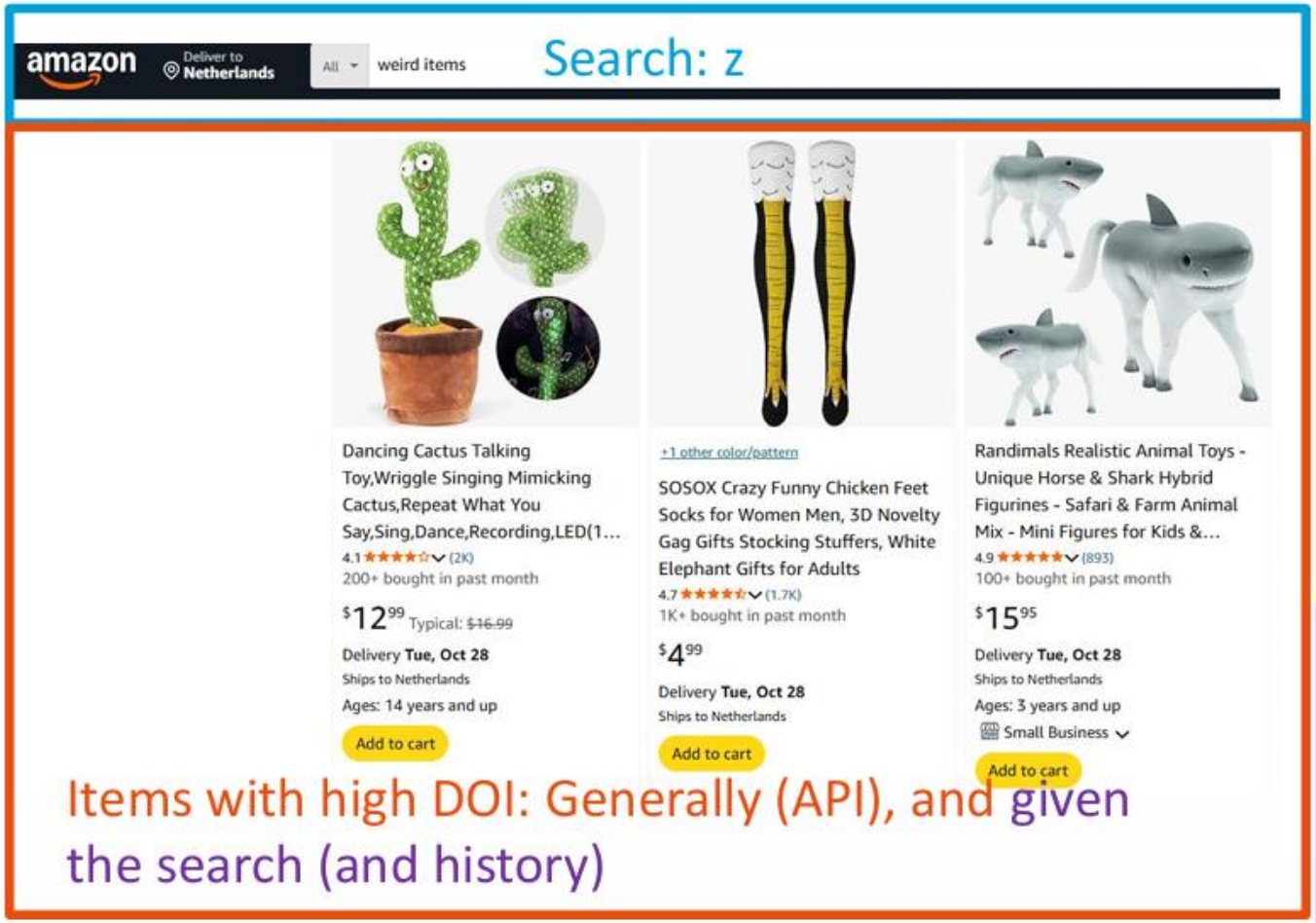


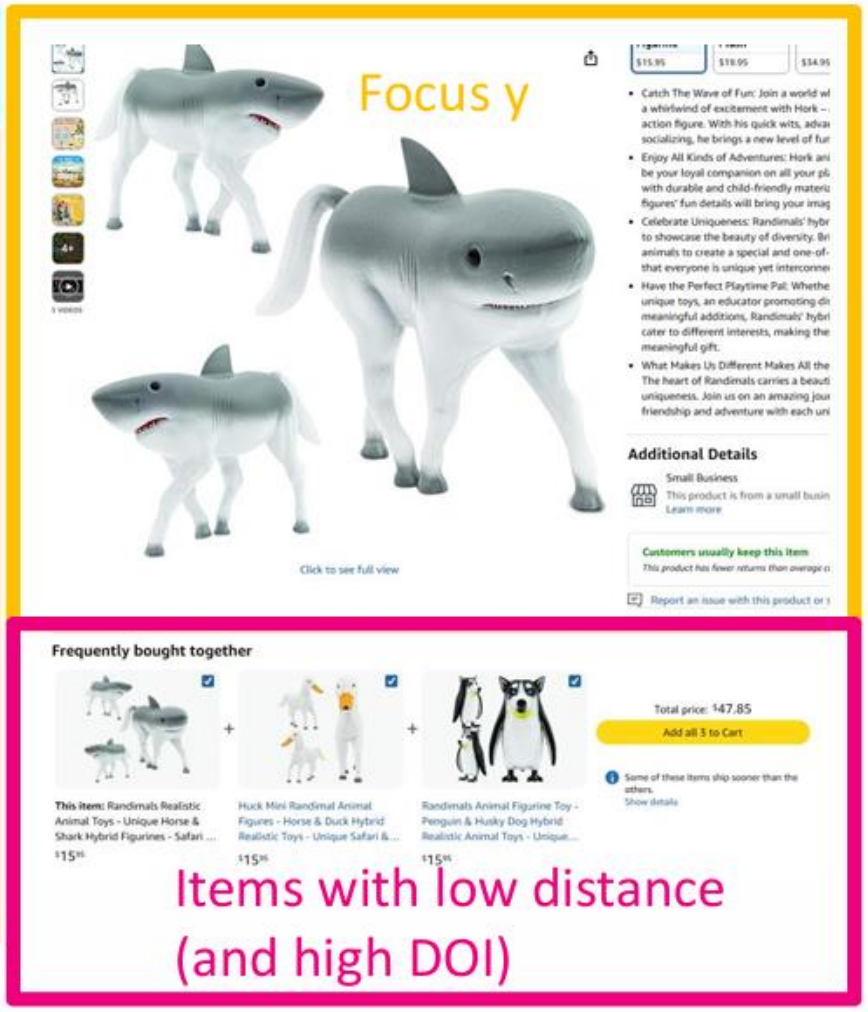
'best' centrality measure is use-case dependent!

Degree-Of-Interest (DOI)

"Search, Show Context, Expand on Demand"

$$\mathrm{DOI}(x \mid y, z) = \alpha \cdot API_{diff}(x) + \beta \cdot UI_{\mathrm{diff}}(x, z) + \gamma \cdot \mathrm{D}(x, y)$$





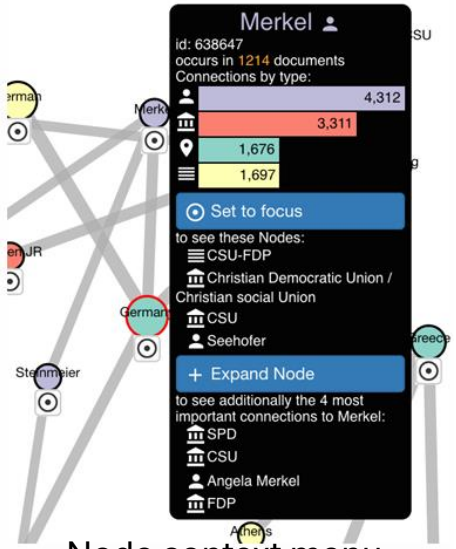
Degree-Of-Interest (DOI)

"Search, Show Context, Expand on Demand"

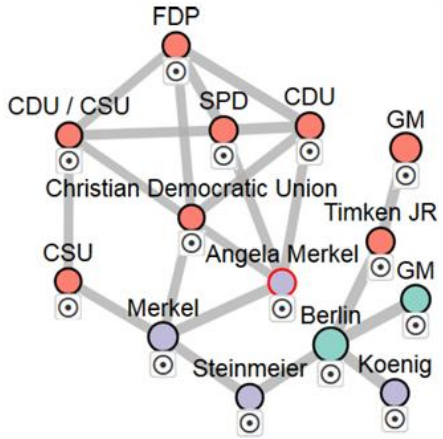
- Edge-Based DOI function:

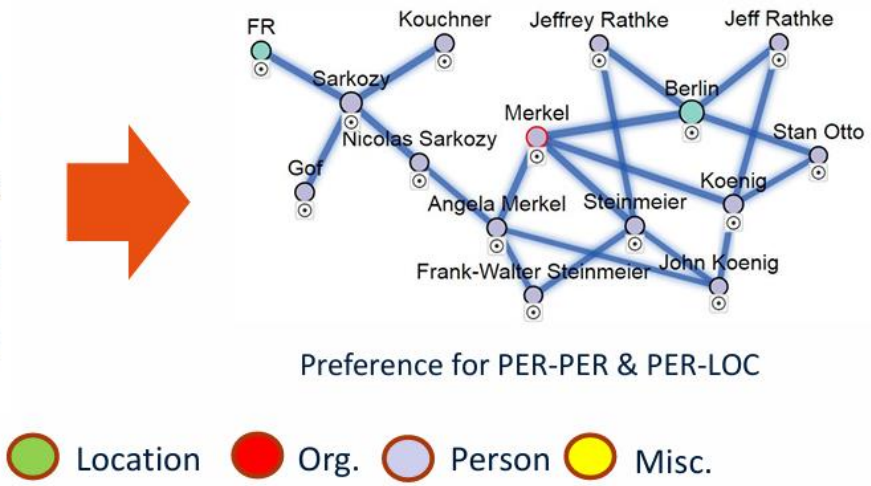
$$DOI(\{u,v\} \mid y,z) = \alpha \cdot API(\{u,v\}) + \beta \cdot UI_{type}(\{u,v\},z) + \kappa \cdot UI_{his}(\{u,v\},y,z) + \gamma \cdot D(\{u,v\},y)$$

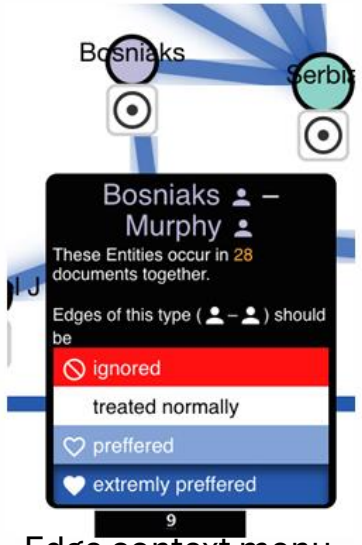
- Here the interest in a node x is replaced by the interest in an edge $\{u,v\}$
- UI_{type}: user interest in entity type
- $UI_{his}(\{u,v\},y,z) = DOI^{old}(x,y \mid y^{old},z^{old})$: include user history in context



Node context menu







Edge context menu

DOI for Timeseries

$$RES_i = \left(\frac{\int_{t_0}^{t_n} DOI(t)\,dt}{\int_{T_0}^{T_N} DOI(t)\,dt} \right)^{1/s},$$

Area in partition

Total area

s is a non-zero scaling factor
s>1: smaller resolution levels are emphasized

s<1: smaller resolution levels are de-emphasized

- Significant object detection for time series → **Multi-Resolution Techniques**

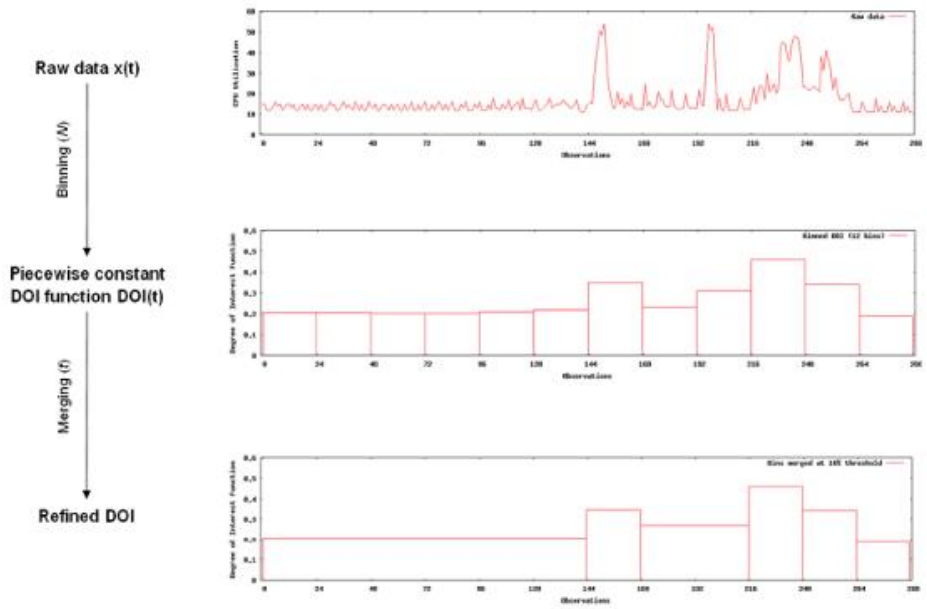
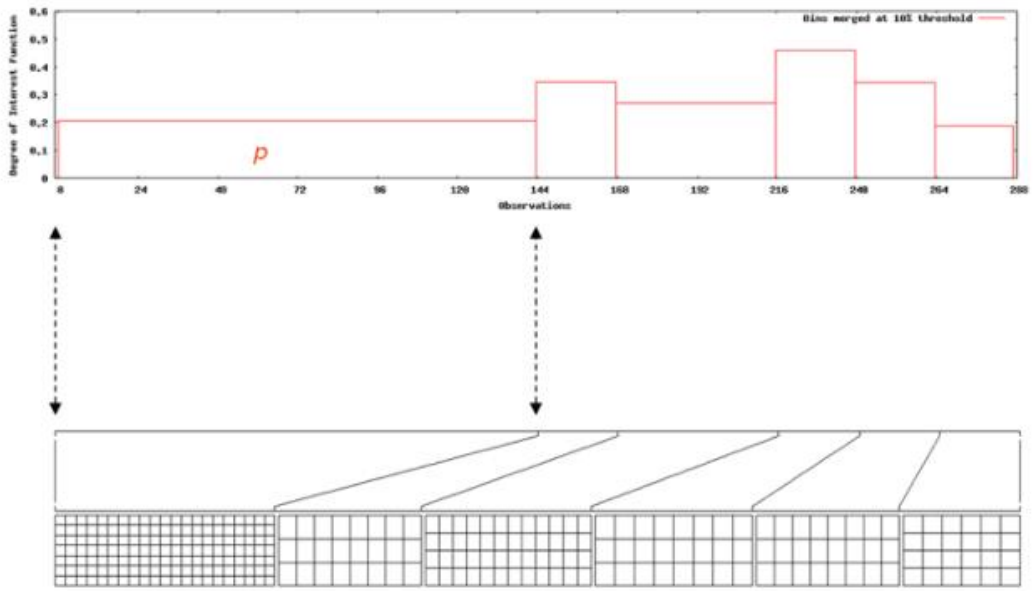
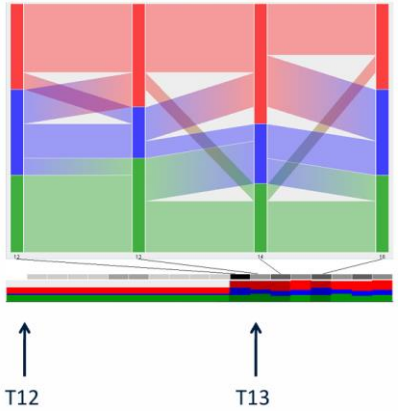


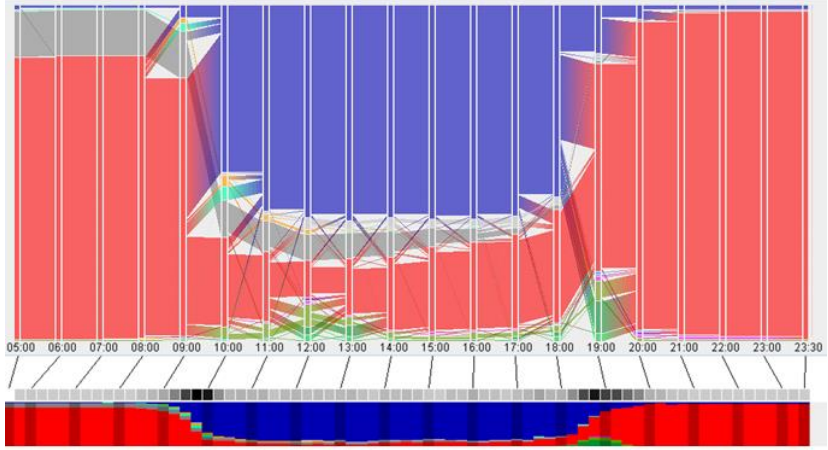
Figure 5: DOI extraction from raw time series by determining the interestingness score over the time series for a number of bins N, followed by (optional) merging of bins with similar interestingness scores.



DOI for Timeseries

- Significant object detection for time series → **Cluster-Based Techniques**
- Example: monthly weather data from multiple stations
  - Create clusters for weather conditions
  - Use dimensionality reduction to map clusters to color space
  - Station s belongs to cluster c at time t:
- Edges represent group changes (parallel sets)
- DOI function: count how many items change cluster per timestep



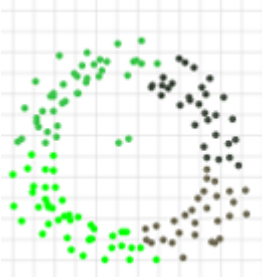


Outlier Detection

- Most straightforward way to detect outliers: **Linkage Clustering**

1. Reduce dimension to a ‘reasonable amount’
2. Iteratively assign objects to the closest cluster
   - (if the distance is more than a certain threshold, create a new cluster)
3. After the clustering is done, check the distances between the clusters:
   - Mark clusters with statistically significant large distances to other clusters as outliers

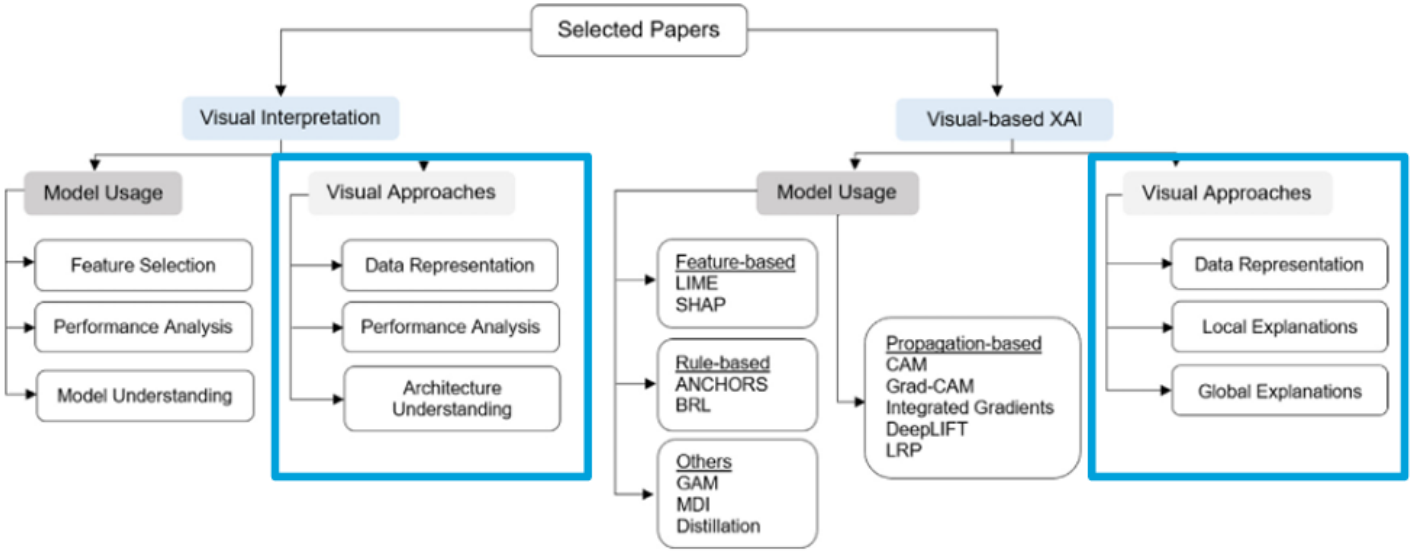
- Can also be combined with most centrality measures:
  - Degree Centrality: encode as adjacency matrix
  - Betweenness Centrality: encode the shortest path distance from u to v as columns
- Not all outliers can be detected through distance-based detection!



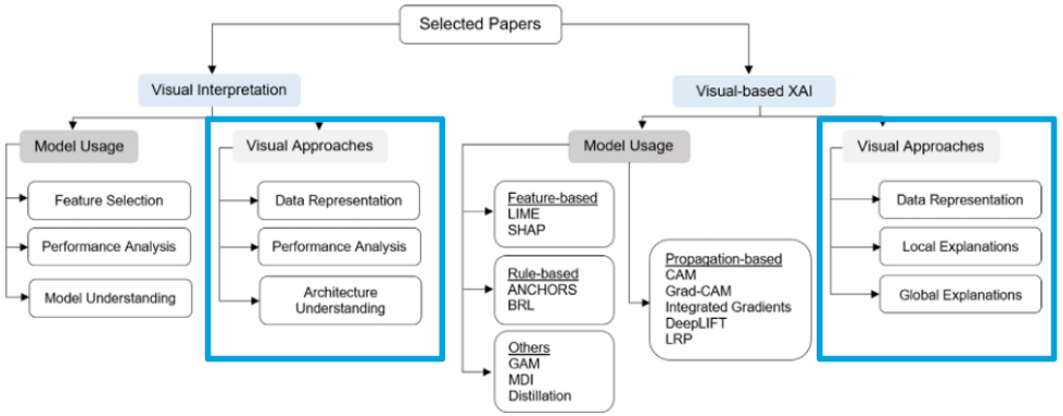
Explainability

- **Explainability**: understanding how a system came up with a given result
- **Interpretability**: understanding how the underlying technology works

- **Visual Interpretation**: techniques to explain models *without* XAI tools
- **Visual-Based XAI**: XAI techniques included in the approach



Explainability



- **Data Representation**: what the model *sees*
  - *Scatter plots, histograms, etc.* to reveal distribution, class imbalance, correlations, etc.
  - *Dimensionality reduction* to show clusters or separability between classes
- **Performance Analysis**: how the model *behaves*
  - *Confusion matrices, Residual plots, etc.*
- **Architecture Understanding**: how the model is *built*
  - *Node-link diagrams* for showing network topology
  - *Visualizations* of intermediate activations
- **Local Explanations**: explain instance decisions (explainability)
- **Global Explanations**: explain decision making process (interpretability)